

Feeds on the Adafruit IoT Dashboard

Table 3

FEED	Work/Process/Activity	Block Title (optional)	Gauge Min. Value	Gauge Max. Value	Gauge Width	Gauge Label	Low Warning Value	High Warning Value	Decimal Places	Test Value
MS1	Soil Sensor 1 Reading	SOIL SENSOR 1	0	1024	25px	Value	350	900	2	45
MS2	Soil Sensor 1 Reading	SOIL SENSOR 2	0	1024	25px	Value	350	900	2	45
MS3	Soil Sensor 1 Reading	SOIL SENSOR 3	0	1024	25px	Value	350	900	2	45
MS4	Soil Sensor 1 Reading	SOIL SENSOR 4	0	1024	25px	Value	350	900	2	45
Tmp	Temp In Degree Celsius	TEMP	10	60	25px	Value	10	40	2	45
Hum	Humidity in %	HUMIDITY	0	100	25px	Value	-	-	2	45
LHT	Light Intensity	LIGHT INTENSITY	0	100	25px	Value	-	-	2	45

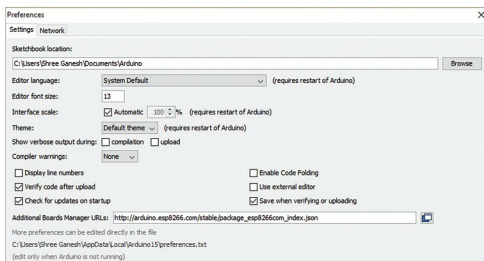


Fig. 6: Software configuration in Arduino IDE



Fig. 7: Adafruit.io dashboard with live data streaming of various sensors outputs

PARTS LIST

Board1	- NodeMCU module
SSI-SS4	- Soil moisture sensor module
DHT11	- Temperature and humidity sensor module
VS1	- Vibration sensor module
RM1	- 5V, 2-channel relay module
	- LDR module
	- 74HC4067 module
	- 5V regulated power supply

```
// Your WIFI password CHANGE IT(2)
#define AIO_SERVER "io.adafruit.com" // Adafruit Server
#define AIO_SERVERPORT 1883
#define AIO_USERNAME "jitendra41085" // Adafruit Username
CHANGE IT (3)
```

```
#define AIO_KEY
"859813f4cb144ee5b0f9e124a239a3f4"
// adafruit Auth KeyCHANGE IT (4)
```

Next, change the software configuration by adding http://arduino.esp8266.com/stable/package_esp8266com_index.json under File/Preferences in Arduino IDE as shown in Fig. 6. Now you can save the code, compile it again, and upload it to NodeMCU board.

After making all the circuit connections, switch on NodeMCU board and open the Adafruit IoT dashboard. You will see something as per Fig. 7. **EFY**

EFY Note
The source code of this project is available for free download at source.efymag.com

Wi-Fi credentials including SSID, passwords, username, and key as shown below.

```
#define WLAN_SSID "Samsungi"
/*Your WIFI NAME SSIDCHANGE IT(1)
#define WLAN_PASS "abcd123456"
```



Jitendra Jangir is training officer (Electronics) at National Skill Training Institute (W), Patna, Bihar



V.K. Shukla is retired as director at Directorate General of Training, Ministry of Skill Development and Entrepreneurship, government of India